

QUESNEL, BC, Canada

WMO: 717790

Lat: 53.0269N

Lon: 122.5064W

Elev: 545

StdP: 94.94

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.6	-23.2	-31.9	0.2	-27.4	-27.3	0.3	-22.5	8.6	-4.5	7.5	-4.6	0.4	0	0.591

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.1	29.9	16.9	27.6	16.1	25.6	15.4	18.1	26.8	17.0	25.2	16.1	23.7	2.2	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.0	11.4	19.7	14.1	10.7	18.7	13.2	10.1	17.9	53.0	26.8	49.7	25.2	47.0	24.0	22.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.5	6.4	5.5	DB	-30.4	33.3	4.6	2.0	-33.7	34.7	-36.4	35.8	-39.0	36.9	-42.4	38.3	(4)
(5)			WB	-30.6	20.2	4.6	1.1	-33.9	21.0	-36.5	21.6	-39.1	22.2	-42.4	23.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.5	-6.3	-4.4	0.8	6.0	11.0	14.5	17.0	16.3	11.5	5.5	-0.8	-6.0
	DBStd	9.70	8.04	6.88	5.57	3.55	3.57	2.81	2.78	2.93	3.43	3.82	5.71	7.11
	HDD10.0	2343	507	403	284	128	31	1	0	0	23	147	323	497
	HDD18.3	4728	765	636	542	370	227	119	61	77	205	398	573	756
	CDD10.0	694	0	0	0	8	63	137	216	196	68	7	1	0
	CDD18.3	37	0	0	0	0	1	4	18	13	1	0	0	0
	CDH23.3	998	0	0	0	3	55	155	408	324	53	0	0	0
	CDH26.7	288	0	0	0	0	8	39	139	91	11	0	0	0
Wind	WSAvg	1.8	1.8	1.8	2.1	2.1	1.9	1.6	1.5	1.4	1.4	1.8	2.0	1.9
Precipitation	PrecAvg	556	50	25	25	31	41	65	66	47	49	55	53	46
	PrecMax	709	93	48	48	51	76	137	112	82	104	89	93	106
	PrecMin	435	25	3	7	9	17	31	21	17	12	20	23	8
	PrecStd	74	21	12	10	10	17	27	26	19	22	20	21	23
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.4	9.5	15.4	23.1	27.6	30.6	33.5	31.5	28.2	19.4	13.8	8.1
		MCWB	4.3	4.3	7.7	11.7	14.4	16.7	17.9	17.3	15.7	12.2	8.1	3.9
	2%	DB	6.4	7.3	12.6	19.0	24.7	27.2	30.3	29.2	24.6	16.2	10.5	5.9
		MCWB	3.0	3.2	6.0	9.8	13.3	15.2	17.1	16.9	15.0	10.2	6.4	2.6
	5%	DB	4.8	5.5	10.6	16.2	22.3	24.8	27.8	27.0	21.4	13.6	7.9	4.1
		MCWB	2.1	2.0	4.8	8.1	12.5	14.5	16.5	16.2	13.9	8.9	4.9	1.6
	10%	DB	2.8	3.8	8.6	13.8	19.8	22.5	25.4	24.7	18.5	11.7	6.1	2.5
		MCWB	1.1	1.1	3.8	6.7	11.4	13.7	15.7	15.5	12.3	7.8	3.6	0.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.8	5.1	8.5	12.0	15.7	18.0	19.5	19.5	16.9	13.1	8.7	4.6
		MCDB	7.7	8.4	14.4	21.2	25.2	27.5	29.9	27.8	24.9	17.8	12.4	7.1
	2%	WB	3.3	3.6	6.5	10.3	14.2	16.4	18.2	18.1	15.6	11.1	6.9	3.0
		MCDB	5.8	6.6	11.5	17.7	22.5	24.7	27.5	26.0	22.8	14.8	9.8	5.4
	5%	WB	2.2	2.5	5.2	8.8	13.1	15.2	17.1	16.9	14.4	9.5	5.3	1.8
		MCDB	4.4	5.0	9.8	14.9	20.5	22.5	25.4	25.1	20.4	12.7	7.5	3.6
	10%	WB	1.1	1.4	4.1	7.4	12.1	14.3	16.2	15.9	13.1	8.1	3.9	0.7
		MCDB	2.6	3.5	8.1	12.7	18.7	20.8	23.7	23.3	17.7	11.2	5.7	2.3
Mean Daily Temperature Range	5% DB	MDBR	7.3	9.4	11.2	13.3	14.2	13.4	14.1	14.8	13.1	10.0	6.6	6.8
		MCDBR	7.9	10.1	13.6	17.8	19.7	19.0	20.1	19.8	18.5	13.4	9.0	7.5
		MCWBR	5.6	6.9	8.1	9.8	9.6	8.5	8.5	8.8	9.8	8.2	6.0	5.5
	5% WB	MCDBR	7.4	9.1	12.9	15.8	17.1	16.4	17.5	18.1	16.8	12.1	8.2	6.9
		MCWBR	5.6	6.5	8.0	8.9	8.7	7.7	7.9	8.5	9.2	7.8	5.8	5.3
Clear-Sky Solar Irradiance	taub		0.269	0.277	0.298	0.328	0.350	0.358	0.348	0.353	0.318	0.300	0.285	0.266
	taud		2.361	2.402	2.452	2.432	2.419	2.413	2.486	2.459	2.564	2.582	2.471	2.330
	Ebn at Noon		732	847	900	908	900	893	896	872	870	817	713	657
	Edh at Noon		56	74	87	101	108	110	100	98	78	62	51	48
All-Sky Solar Radiation	RadAvg		0.85	1.82	3.26	4.71	5.42	5.90	5.90	5.03	3.42	1.97	0.92	0.64
	RadStd		0.08	0.19	0.22	0.31	0.38	0.44	0.55	0.22	0.29	0.17	0.09	0.06

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	-0.28	N/A	N/A	N/A	+76	N/A
(47)	Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page